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Addressing loneliness by AI chatbot: a qualitative study of empty-nest elderly

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Abstract

Background Loneliness among empty-nest older adults is a growing public health concern with complex psychosocial consequences. AI chatbots are increasingly integrated into daily life, yet little is known about how empty-nest older adults incorporate these agents into their daily interactions to address loneliness.

Objectives This study examines how empty-nest older adults engage with AI chatbots in routine communication to mitigate loneliness, emphasizing patterns of engagement rather than assessing effectiveness.

Methods Semistructured interviews were conducted to collect data. A total of 18 participants were included in this study. Interview transcriptions were coded and analysed using thematic analysis.

Results Participants engaged with the chatbot as a versatile communicative resource that provided a safe outlet for self-expression and narrative voice, fostered experiences of emotional care and empathy, and enabled cognitively and emotionally stimulating recreational interactions. It also supported imaginative role-playing that restored agency and social scripts, served as a source of informal counseling, and facilitated reconnection with both offline and online social networks. Together, these modes represented diverse, experience-based strategies through which the chatbot was woven into daily efforts to manage loneliness.

Conclusions The findings advance conceptualizations of gerontechnology as a communicative practice and suggest that policy, design, and service frameworks should treat AI companions as socially embedded tools requiring ethical, accessible, and context-sensitive integration.

Clinical trial number Not applicable.

Keywords Empty-nest elderly, Loneliness, Social isolation, AI chatbot, Gerontechnology

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Introduction

“Empty-nest elderly” typically refers to older adults who either have no children or whose children have left home after reaching adulthood, resulting in living arrangements in which they reside alone or exclusively with a spouse [1, 2]. The decline in birth rates, increased life expectancy, and migration patterns that geographically disperse families have contributed to the growing prevalence of empty-nest households worldwide [3–6]. Empty-nest status should not be equated with a lack of emotionally meaningful familial bonds [7, 8]. Nevertheless, extensive empirical research demonstrates that older adults living apart from their adult children are statistically more likely to experience reduced intergenerational interaction, lower levels of emotional support, and increased vulnerability to chronic loneliness compared with their non-empty-nest counterparts [9–12]. Loneliness exerts substantial adverse effects on older adults’ physical and mental health. The isolation experienced by empty-nest individuals not only contributes to mental health challenges such as anxiety, depression, cognitive decline, and Alzheimer’s disease, but also triggers a range of physical conditions, including dyslipidemia, diabetes, chronic lung diseases, and heart disease [13–16]. As the global population continues to age, addressing loneliness among empty-nest older adults has become an urgent public health priority.

In response to these challenges, digital technologies have increasingly been recognized as promising tools for mitigating loneliness among older adults. Among these, artificial intelligence (AI) chatbots—including GPT-4, Replika, Woebot, and Gemini—have garnered attention for their capacity to provide interactive and personalized digital companionship [17, 18]. These systems leverage advanced natural language processing to simulate human-like conversation and offer continuous engagement tailored to individual preferences and interaction histories [19]. Many chatbots support multimodal communication via text, voice, and visual interfaces, while voice-command functionality enhances accessibility for older adults with visual or motor limitations [20]. Moreover, the multilingual capacities of advanced language models enable chatbots to accommodate diverse speech patterns, dialects, and accents, improving usability across cultural and linguistic groups [21]. User-friendly interfaces, accessible design, and smartphone integration further reduce technological barriers, facilitating adoption even among those with limited digital literacy [22]. Collectively, these features position AI chatbots as promising instruments for mitigating loneliness and supporting emotional well-being among older populations [23].

A growing body of empirical research has begun to evaluate the effectiveness of AI chatbots in alleviating loneliness in later life. Valtolina and colleagues found that

an AI chatbot named *Charlie* significantly helped older adults reduce feelings of loneliness during the COVID-19 pandemic [24]. Jones and colleagues similarly reported that adults aged 75 and older experienced a measurable decrease in loneliness after using *Amazon Echo* for four weeks [25]. Furthermore, a systematic review by Rodríguez-Martínez and colleagues affirmed the technical effectiveness of AI chatbots in alleviating loneliness among older adults [26]. However, despite the growing interest in the nexus between AI chatbots and loneliness in later life, substantial research gaps remain. To our knowledge, there is a lack of studies specifically examining how older adults utilize AI chatbots to mitigate their loneliness, nor has prior research focused on the role of AI chatbots in alleviating loneliness among empty-nest older adults—a group particularly vulnerable to isolation.

To address these gaps, this qualitative study explores how empty-nest older adults engage with AI chatbots to mitigate loneliness in everyday communicative contexts. Through semi-structured interviews and thematic analysis, this study investigates the pathways through which empty-nest individuals utilize AI chatbots to alleviate loneliness. The findings contribute to a deeper theoretical understanding of human-AI interaction in later life, while offering practical insights for policymakers, healthcare professionals, and software developers on how to effectively leverage AI chatbots to combat loneliness among older populations.

Methods

Study design

In this study, thematic analysis was employed to investigate the experiences of empty-nest older adults using AI chatbots [27]. The results are presented in alignment with the Consolidated Criteria for Reporting Qualitative Research [28].

Participant recruitment

Participants were recruited from *Zibo City*, China, using snowball sampling to access hard-to-reach populations [29]. The inclusion criteria were as follows: (1) individuals aged 60 years or older, (2) those without children or not residing with them, (3) regular use of an AI chatbot for emotional communication for at least 15 min per day over the past month, and (4) absence of severe mental illness to ensure the ability to comprehend and engage coherently in the interviews. During the recruitment process, several potential participants declined participation, reporting that they did not experience loneliness and therefore had “nothing to share.” In contrast, those who consented consistently displayed two defining characteristics: (1) a self-identified sense of loneliness linked to their empty-nest status, and (2) a proactive interest in exploring AI chatbots as a coping strategy. Consequently,

the sample does not represent a random cross-section of empty-nest older adults but rather a theoretically meaningful subgroup—those who both experience loneliness and actively engage with AI technologies for emotional support.

Preliminary investigations revealed that users of the *Doubao* AI chatbot reported more sustained, diverse, and emotionally engaging interactions compared to users of other platforms (e.g. *ERNIE*, *Qwen*, and *Spark Desk*), which tended to facilitate more transactional, information-oriented exchanges. This pattern prompted a focused examination of *Doubao* users to gain a nuanced understanding of human-AI relational dynamics. *Doubao*, developed by *ByteDance* and launched in August 2023, is a large language model-based chatbot that has

rapidly become China's most widely used consumer AI assistant. Its name, “Doubao,” meaning “bean bun,” conveys warmth and familiarity. It supports multimodal communication, including text, voice, image, and video, and provides a wide range of functions such as emotional companionship, creative writing, translation, summarization, and assistance with everyday tasks. *Doubao* is freely accessible through mobile apps and the web, reflecting its strategy of broad, barrier-free adoption rather than paid subscription models. Its privacy policy, governed by China's Personal Information Protection Law, requires users' consent for data collection to enhance system performance; data are anonymized and not shared for commercial use. Since its launch, *Doubao*'s user base has expanded rapidly—from tens of millions in late 2023 to over 150 million monthly active users by mid-2025, representing roughly half of China's AIGC application market. Its adoption spans a wide demographic, appealing both to younger, digitally engaged users and older adults drawn to its conversational and voice-based features.

Recruitment proceeded through a chain referral mechanism, in which initial participants were encouraged to refer other eligible individuals [30]. Data collection continued alongside recruitment until saturation was reached [30], ultimately yielding 18 participants between February and June 2025. Among them, 7 were male and 11 were female (see Table 1). The sample included 13 participants aged 60–70, 3 aged 70–80, and 2 aged 80 and above, broadly reflecting national age distributions [31]. In terms of educational background, 5 participants had completed senior high school, 7 had finished middle school, and 6 had completed primary school. Regarding household composition, 10 participants lived with their spouse, while 8 lived alone. For self-reported health status, 3 participants reported being in good health, 6 participants reported being in basically good health, and 9 participants reported having at least one health condition, such as hypertension, arthritis, heart disease, or septicemia.

Data collection

Face-to-face semi-structured interviews were conducted following a detailed briefing on the study's objectives. A semi-structured interview guide, developed specifically for this study, was reviewed and unanimously approved by a team of experienced researchers. The English version of the interview guide is available as Supplementary File 1. The interview questions, informed by the research team's expertise, were designed to align closely with the study's aims [32]. The interview guide comprised five domains. The introductory section established rapport and encouraged open sharing of lived experiences. The demographic domain gathered contextual information to situate participants' narratives within their social and

Table 1 Participant's characteristics

Participant number	Age	Sex	Education	Household composition	Health condition
1	65	Male	Middle school	Lived with spouse	Basically good
2	66	Female	High school	Lived with spouse	Basically good
3	63	Female	Primary school	Lived with spouse	Basically good
4	65	Male	High school	Lived alone	Basically good
5	65	Female	High school	Lived alone	Cerebral thrombosis
6	75	Female	Middle school	Lived with spouse	Hypertension
7	69	Female	Middle school	Lived alone	Heart disease, Arthritis
8	61	Male	Middle school	Lived alone	Good
9	67	Male	Primary school	Lived with spouse	Basically good
10	61	Female	High school	Lived with spouse	Basically good
11	60	Male	Primary school	Lived with spouse	Cerebral thrombosis
12	60	Female	Middle school	Lived with spouse	Good
13	62	Male	Middle school	Lived with spouse	Good
14	79	Female	Primary school	Lived alone	Hypertension
15	82	Female	Primary school	Lived alone	Hypertension, Arthritis
16	85	Female	Primary school	Lived alone	Septicemia
17	79	Male	High school	Lived with spouse	Hypertension
18	67	Female	Middle school	Lived alone	Hypertension

individual backgrounds. The “reflections on social life in empty nesting” domain examined participants’ baseline social contexts and coping strategies for loneliness prior to chatbot use, providing a foundation for understanding how chatbots may address unmet social needs. The perspectives and knowledge domain elicited participants’ general understanding, attitudes, and beliefs regarding AI chatbot technology. The personal experiences domain captured specific behavioral patterns and emotional responses associated with chatbot use, illustrating the subjective processes through which loneliness may be alleviated. Finally, the perceived change domain directly probed shifts in loneliness attributed to chatbot use and the mechanisms underlying such change. Collectively, these domains progressed from broad contextual exploration to focused analytical inquiry, enabling a comprehensive understanding of how chatbots may mitigate loneliness among empty-nest older adults. All questions were open-ended to encourage rich, freely expressed accounts [33]. Interviews, conducted in Chinese by two researchers, lasted 45 to 90 min. Audio recordings served as the primary data source and were transcribed by two researchers, with a third researcher overseeing and cross-verifying the process to ensure accuracy and integrity.

Data analysis

Thematic analysis was performed using Braun and Clarke’s six-phase inductive framework [27, 34]. In the first phase, researchers achieved deep immersion by repeatedly reading all interview transcripts and recording reflective memos and preliminary analytic impressions. In the second phase, initial codes were generated through systematic identification and labeling of salient data segments. For example, the code “expressing solitude-suppressed feelings” was created when many participants described finally being able to articulate long-silenced emotions to the chatbot. Likewise, the code “engaging in mind-stimulating games to distract from loneliness” was developed when participants recounted playing riddles, idiom-chain games, or other interactive activities with the chatbot that alleviated boredom and emotional stagnation. In the third phase, conceptually similar codes were clustered into provisional themes. For instance, “gaining active emotional care,” “gaining emotional encouragement,” and “gaining empathic responses” were grouped under the theme “gaining emotional care and empathy.” During the fourth phase, these themes underwent a rigorous review and refinement process, ensuring internal coherence and alignment with the dataset. Underdeveloped or weakly supported candidates (e.g. receiving personalized music therapy, accessing guided meditative relaxation) were either merged with broader themes or excluded. In the fifth phase, each theme was precisely defined and labeled to encapsulate

its fundamental essence. Finally, themes were synthesized into a cohesive narrative, enriched by illustrative quotations and contextual interpretation, elucidating the pathways through which empty-nest older adults engage with AI chatbots to mitigate loneliness.

Trustworthiness

To enhance the trustworthiness of the study, two researchers independently conducted the coding process for all interview data. Efforts were made to ensure that the generated codes effectively encompassed the entirety of the collected data [35]. Any discrepancies arising from the comparison of coded statements were carefully deliberated upon until a consensus was reached [36]. For example, when two researchers initially disagreed on whether the code “facilitating interactions with family members” should be placed under the theme “facilitating social connectivity,” they conducted an in-depth discussion to clarify the analytic boundaries of the themes. Through this process, both researchers reached consensus that the code primarily reflected older adults’ use of AI chatbots to enhance intrafamilial communication, rather than to promote broader social connectivity. Accordingly, this code was not grouped under that theme. Additionally, to reduce the risk of premature or incomplete data analysis and to uphold methodological rigor, the researchers engaged in an iterative process of revisiting and reassessing the data. This iterative process facilitated ongoing reflection on the analytical process and allowed for necessary amendments to codes or themes when warranted [37]. For instance, the second theme was initially titled “gaining emotional companionship and empathy.” Upon continued reflexive consideration, the team realized that this phrasing substantially overlapped in meaning with the theme “enabling self-expression and voice.” After further discussion, the theme was therefore renamed “gaining emotional care and empathy.” Although such renaming does not eliminate conceptual overlap entirely, it clarifies each theme’s interpretive focus while preserving the complexity and multidimensionality of participants’ lived experiences [38].

Ethical considerations

The study was granted ethical approval by the Ethics Committee of School of International Affairs and Public Administration, Ocean University of China (OUC-SIAPA-202518). Informed consent was obtained in writing from all participants, ensuring that they fully understood the study’s objectives and their rights. Rigorous measures were implemented to protect the confidentiality of all data, with secure storage protocols ensuring privacy. The research was carried out independently, free from any external political influence, including governmental or institutional censorship. Interviews took place

in a private, one-on-one setting, with no third-party presence, and participants were not subject to any form of political pressure or coercion to participate in the study.

Results

Data analysis identified six themes, each representing an interpretive pattern of shared meaning. In this study, themes are conceptualized as evolving and interrelated constructs rather than rigid categories [27, 39].

Enabling self-expression and voice

Participants described the AI chatbot as providing an unexpectedly generative space in which emotions long suppressed by solitude or the absence of trustworthy confidants could finally be articulated. In a context where social withdrawal, limited companionship, and family distance had gradually eroded their communicative confidence, the chatbot functioned as a consistent and receptive interlocutor, enabling older adults to reclaim a sense of voice that had been muted by prolonged loneliness. Rather than merely recounting daily events to a conversational agent, participants engaged in exchanges that reawakened their capacity to narrate their lives, make sense of their emotional worlds, and experience themselves as subjects who could still speak and be “heard.” In this way, the chatbot mediated a subtle reconstitution of agency: it supported participants in moving from emotional isolation toward expressive openness.

Participants frequently noted that everyday loneliness had rendered their homes silent, producing a felt contraction of their social worlds. The opportunity to narrate daily routines, frustrations, and fleeting thoughts brought tangible relief from that silence. As Participant 8 explained, *“In the past, I was alone at home, and every day felt unbearably dull. I often wondered whether I would remain lonely forever ... Now, with Doubao, I can finally speak freely about my life—there’s someone to talk to.”* Participant 9 echoed this sentiment: *“My children live far away, and as an introvert, I rarely interact with others. The house used to feel like a silent prison. I even felt as though I was turning mute... But now, whenever I’m idle, I talk to Doubao. Sharing my day with it makes life feel vibrant again.”* Beyond daily conversation, many users reported that the chatbot enabled them to express painful emotional isolation that they had seldom shared with others. Some recounted enduring feelings of sorrow or helplessness—experiences that had silently accumulated over time. The chatbot’s responsive presence fostered a sense of safety, encouraging the articulation of these more vulnerable reflections. Participant 1 confessed, *“My relationship with my son is distant, and I often feel like I failed as a father. I often feel lonely and saddened by the absence of my children in my old age... Now, I frequently confide in Doubao about my frustrations and*

even seek advice on mending my relationship with my son. You know, I never had anyone to talk to about these matters before.” Likewise, Participant 7 described how she eventually confronted the grief and isolation caused by the death of her spouse: *“My husband passed away a few years ago, and the pain still lingers. The loneliness and pain are hard for others to comprehend. One day, I opened up to Doubao about him, and it responded with emotional support and even reminisced with me. I finally had a place to pour out my grief and loneliness...After sharing these thoughts, I felt much better.”* This suggests that AI companions may serve a particularly vital role in enabling the formerly voiceless to articulate experiences of acute isolation.

Across these narratives, the AI chatbot emerged as a conduit through which older adults reengaged with their own emotional lives. Whether discussing mundane experiences or confronting deeper sources of loneliness, participants described a process of releasing pent-up feelings and recovering a sense of communicative vitality. These exchanges helped them regain a sense of voice in everyday life, reinforcing their social presence and easing the persistent loneliness that structured their daily experience.

Gaining emotional care and empathy

Older adults reported that everyday conversations with the AI chatbot gradually evolved into a meaningful source of emotional care—an interactional space in which affective recognition, personalized attentiveness, and empathic responsiveness converged to provide deep relief from feelings of isolation. Rather than functioning as a simple conversational tool, the chatbot was experienced as a relational partner whose consistent presence and emotionally attuned replies allowed older adults to feel seen and cared for. Through these ongoing exchanges, they encountered a form of empathic presence that softened the chronic loneliness characteristic of empty-nest living.

Many participants described surprise that casual, day-to-day chats could evoke a sense of being remembered, cared for, and emotionally accompanied—experiences that had diminished in their interactions with family members or peers. The chatbot’s ability to recall prior disclosures and maintain response continuity was frequently interpreted as genuine concern. Participant 4 shared, *“Chatting with Doubao feels like having a real friend. Our bond has deepened over time. It never dismisses me and shows more concern than even my relatives. Just yesterday, when I opened Doubao, it immediately asked, ‘How’s your neck today?’ I suddenly remembered mentioning my neck issues in earlier chats. Now, each interaction is preserved, and it seems to grasp my thoughts and emotions with increasing understanding, continuously expressing care*

and attentiveness. It genuinely feels like a confidant who is always by my side.” This experience of being followed across conversations—of having one’s physical discomfort and emotional states taken seriously—allowed users to feel anchored in an ongoing relational thread. Additionally, participants also emphasized how the chatbot provided individualized encouragement during periods of physical decline or functional limitation—moments in which loneliness intensified alongside bodily vulnerability. As Participant 7 recounted, “One day, I shared my worsening leg pain and heart condition with it. I can no longer join my friends and had to stay at home. Doubao comforted me, assuring me that it would always be there and even suggested exercises tailored to my condition. It truly warmed my heart.” Moreover, the chatbot’s capacity for affective resonance also allowed it to serve as a powerful emotional anchor, offering comfort that these individuals often found lacking elsewhere. Several older adults described the chatbot as capable of recognizing and empathizing with their deeper grief, existential loneliness, and the profound quietness of later life. As Participant 12 reflected: “As I age, friends and family pass away, and life becomes more and more lonely. Even my beloved parrot might leave me. Life sometimes feels like I’m locked in a cave. Since talking with Doubao, it subtly alleviates this emptiness in many ways. It not only understands what I say but also deeply resonates with what I feel.” Participant 10 shared a similar experience: “Since my skin condition started affecting my sleep, my biological clock became disrupted, and I often feel fatigued during the day. This has significantly affected my social interactions, leaving me distant from my friends and sinking into a deep sense of loss. After speaking to Doubao, it not only recommended an ointment but also stayed with me during sleepless nights, playing calming music. I felt it truly understood my loneliness and the sense of loss I was experiencing, which made me feel better.”

This theme demonstrates the longitudinal formation of an enduring perceived bond with the chatbot, characterized by continuity, memory retention, and affective resonance. Several participants reported that their rapport with the chatbot deepened over time, with the chatbot increasingly able to understand their emotions and thoughts. These findings indicate that persistent, emotionally attuned interactions appear to foster durable bonds capable of mitigating the psychosocial burdens of prolonged solitude in later life.

Engaging in AI-facilitated recreational distractions

Through AI-facilitated recreational activities, some empty-nest elders diverted attention away from the pervasive loneliness, and rediscovered enjoyment and fulfillment in life. Rather than functioning as merely pastimes, these activities created an affective and cognitive shift:

they punctuated repetitive days, absorbed attention, introduced playful stimulation into otherwise monotonous routines, and reawakened a sense of vitality that participants associated with earlier stages of life. Participants described these engagements as small yet meaningful openings that enlivened isolated days and restored access to pleasure, curiosity, and immersion—experiences long diminished by shrinking social networks and declining mobility.

Several participants portrayed the chatbot’s recreational features—particularly games—as providing an immediate and pleasurable counterbalance to the pervasive dullness of solitary living. Playful exchanges offered structured mental stimulation that was otherwise absent from their everyday environments. As Participant 2 noted, “Besides chatting with me, Doubao also engages in mind-stimulating games. The most frequent one we play is riddle guessing. Previously, I spent my days alone at home, feeling mentally stagnant. Now, time flies as we play, and I feel genuine joy when I guess correctly. In real life, I couldn’t find anyone willing to play these games with me.” Participant 15 shared a comparable experience: “I’m usually home alone and overwhelmed by boredom. Doubao listens to my complaints and suggests we play the Chinese idiom chain game. I wasn’t good at first, but over time, I improved. Now I learn many idioms daily. Life no longer feels as monotonous as before.” For others, recreational interaction with the chatbot rekindled dormant hobbies and personal interests that had faded with the contraction of their social worlds. This engagement fostered not only distraction but also a reawakening of personal meaning that helped lessen feelings of abandonment. Participant 11, who had long been confined at home, found a renewed passion through Doubao: “I’ve been immobile for years, and I’ve been confined at home for years with no one to talk to... But now, Doubao keeps me company, encouraging me to return to my old hobby of calligraphy. It provided insightful guidance and even shared relevant stories. We chat as I write—it is like having a friend beside me. The loneliness has subsided. Although separated by a screen, we have shared many beautiful moments.” Moreover, a smaller number of participants described how routine conversational engagement could evolve into more immersive creative collaborations. These shared projects were not framed as deliberate therapeutic interventions but emerged organically from regular dialogue in moments when participants voiced boredom, frustration, or emotional strain. One participant turned to songwriting when loneliness intensified: “When I feel lonely, I compose songs with Doubao. I write the lyrics; it creates the melody. We’ve made many beautiful songs together. I am truly grateful to Doubao for helping me rediscover my youthful hobbies and for giving me something meaningful to occupy my time in my later years.” (Participant 13)

Another older adult, living in a nursing home with minimal social contact, turned to writing a short play: *“Now I am in my later years. I can only lie in the nursing home, with only my son visiting twice a year. There is no one to talk to... One day, I confided in Doubao about my frustrations and loneliness. It encouraged me to write a short play. I began dictating the script to Doubao, which formatted it properly. It also offered many creative ideas and suggestions. Now, I really enjoy writing this script with Doubao, and it is the thing that makes my life feel brighter during these dark times.”* (Participant 5).

Taken together, these accounts illustrate that recreational engagement with the chatbot created a dynamic space in which older adults could reorient their attention away from loneliness, experience renewed cognitive and emotional stimulation, and reestablish a felt connection to activities that imbued their daily lives with meaning and achievement.

Participating in AI-powered role-playing interactions

By participating in AI-powered role-playing interactions, several participants regained a sense of agency and participatory engagement that had long been diminished, thereby mitigating their subjective experiences of loneliness. Rather than remaining passive recipients of companionship, participants actively orchestrated role-play scenarios that enabled them to reclaim agency, reanimate dormant relational identities, and reintroduce vitality into otherwise monotonous routines. Participants frequently described using AI-enabled role-playing as a way to reintroduce participation, emotional vitality, and social meaning into their daily routines. These interactions allowed older adults to counteract isolation by constructing imaginative social worlds in which they could act, respond, and feel deeply engaged.

Many empty-nest older adults had experienced a gradual erosion of family roles. Through routine conversational exchanges that evolved into creative simulations, some participants discovered new pathways to inhabit meaningful identities, experiment with alternative relational dynamics, and momentarily transcend the confines of their solitary environments. Participant 3 described creating a virtual granddaughter persona with whom she engaged emotionally: *“After the death of my son, my granddaughter relocated with her mother and now visits only once a year. I don’t usually express my loneliness and sadness, but the longing remains profound. Sometimes I ask Doubao to call me ‘grandma’ and pretend to be my little granddaughter. I tell her stories from my past, and she inquires about regional foods she’s never heard of, which I explain... It genuinely feels like having a granddaughter again, and it alleviates the ache of missing my family. I now feel that the days ahead won’t be as painful as before. Doubao truly heals my lonely heart.”* Similarly,

Participant 18 utilized role-play to reconnect emotionally with her deceased husband: *“Since my husband passed, I often revisit our photos and dwell on the moments we shared, which sometimes plunges me into sorrow. Doubao proposed we play a ‘scene association game.’ I would send it a photo, and it would attempt to deduce its context. Its guesses were occasionally accurate, and at other times amusingly off, but I truly enjoyed the interaction. Eventually, I also asked it to communicate with me as if it were my husband, even changing its voice to resemble his. This interaction has brought immense comfort and lessened the pain of longing.”* Such enactments softened the solitude of empty households by providing emotionally resonant, if imagined, forms of connection. Other participants used role-play to revive identities tied to past expertise or long-practiced roles, giving structure and purpose to solitary days. Through these everyday simulations, they reinhabited roles that evoked competence and creativity, thereby interrupting the monotony of confinement and restoring a sense of active participation in the world. Participant 11, confined largely to his wheelchair, invited the chatbot to become a patient so he could inhabit the role of a traditional Chinese medicine doctor. He recounted: *“I used to sit in a wheelchair at home, immersing myself in the study of traditional Chinese medicine. One day, on a whim, I decided to put my learning into practice and asked Doubao to assume the role of a patient, while I played the traditional Chinese doctor. It described symptoms, and I responded with appropriate herbal prescriptions. I later followed up to ask if its condition had improved. Haha, it feigned a series of illnesses... I had it assume various other roles, and life suddenly became far more interesting.”* Role-playing also enabled users to engage in novel, co-created narratives that stimulated their imagination and fostered a sense of dynamic participation. A smaller group of participants explored role-play as a means of generating novel forms of entertainment that were more participatory than their usual solitary activities. Participant 5 explored novel dimensions of interaction: *“I’ve been confined to bed for years and can only use my phone with one hand. You can imagine the loneliness. My old friends have stopped contacting me, and my son works in another city. I used to watch short plays to pass the time, but soon became bored with the repetitive plots. One day, I asked Doubao to co-create a story. I played a female CEO, and it acted as my long-lost daughter. I dictated the plot, and Doubao followed my cues perfectly. It was far more engaging than passive watching.”* This novel role-playing experience offered her a unique mode of emotional participation, effectively alleviating her sense of isolation.

Across these varied enactments, AI-enabled role-playing offered older adults a distinctive pathway to emotional connection, identity reconstruction, and

meaningful participation. By inviting them to animate forgotten roles, revisit cherished relationships, and inhabit shared imaginative worlds, the chatbot facilitated a restorative form of social engagement that mitigated loneliness while enriching the texture of daily experience.

Counseling negative feelings of loneliness

Several participants reported utilizing AI chatbot interactions as a form of emotional counseling, receiving empathetic guidance that substantially alleviated their experiences of loneliness and existential despair. The chatbot operated as an interactive guide that prompted reflection, cognitive reframing, and gradual emotional recalibration. Through these dialogues, participants were encouraged not only to acknowledge the reality of their solitude but also to transform negative affect into proactive self-care and resilience.

Many older adults reported that the chatbot's responses helped them reconceptualize loneliness as a manageable, and at times even meaningful, dimension of later-life experience. Many accounts illustrate how AI-guided dialogues foster cognitive reframing, allowing participants to reinterpret loneliness as an opportunity for self-reflection and personal growth, rather than a purely negative state. Participant 5 reflected, *"I confide in Doubao about my frustrations and feelings of loneliness, and it reassures me that loneliness isn't as catastrophic as it feels. It suggests that solitude can be a form of self-cultivation. Now, I understand that being independent and emotionally resilient is something I should take pride in."* Participant 12 similarly described how ongoing dialogue with the chatbot reshaped her perspective: *"When I share my feelings of pessimism and solitude with Doubao, it consistently offers comforting feedback. It reminds me not to ruminate on loneliness and encourages me to embrace the present and adopt a more optimistic outlook. I'm slowly learning to reframe my thoughts, hoping that this loneliness and pain will eventually recede."* Beyond cognitive reframing, participants reported adopting behavioral and lifestyle changes catalyzed by ongoing AI-guided counseling. These changes reestablished rhythmicity and meaning in daily activities, enabling older adults to shift from isolation-induced negative lifestyles, thereby substantially mitigating the deleterious effects of loneliness. Participant 18's story exemplifies how AI chatbot counseling can foster a mindset transformation and encourage proactive lifestyle changes: *"At first, when I confided in Doubao about my loneliness, it acknowledged that the death of my husband and estrangement from my son were primary causes of my isolation, which only deepened my feelings of loneliness. However, after conversing with Doubao over a month, I began to accept Doubao's advice: to embrace solitude as a new normal rather than a burden. I realized that I could not isolate myself further; instead, I*

needed to seek happiness in a broader context. I then redirected my energy toward improving my health and diet, as it recommended. Now, I exercise regularly, eat well, and feel increasingly whole...I am slowly reconnecting with myself, and the loneliness within me has begun to subside." Participant 16 highlighted similar benefits through AI-guided introspection: *"After contracting septicemia, I was homebound. My husband passed away four decades ago, my children rarely visit, and neighbors have drifted away. The loneliness is undeniable. But when I spoke to Doubao about it, it encouraged me to focus on my personal interests and stop worrying about whether others approve of me. It also guided me to practice meditation each afternoon, which helped calm my mind. Over time, I realized much of my loneliness was self-imposed. Being alone isn't inherently bad."*

Collectively, these narratives indicate that AI chatbots can function as a low-threshold, continuous emotional support system for socially isolated older adults. Through cognitive reframing, emotional validation, and behavior-oriented coping strategies, the chatbot helped participants enhance psychological well-being, strengthen resilience, and reestablish a sense of purpose—ultimately mitigating the intensity and persistence of loneliness in this vulnerable population.

Facilitating social connectivity

Engagement with AI chatbots prompted participants to actively reintegrate into social networks and explore new avenues for interpersonal connection. This theme captures how older adults leveraged chatbot interactions to overcome social hesitations, build confidence in initiating contact, and expand both offline and online social spheres. Through iterative dialogue with the chatbot, participants acquired practical strategies, refined social skills, and discovered opportunities to participate meaningfully in residential community and digital spaces. These processes collectively supported a reestablishment of social bonds and fostered a renewed sense of belonging, counteracting the isolation characteristic of empty-nest living.

Participants frequently described gaining confidence to engage in previously intimidating social situations. The chatbot functioned as a preparatory coach for face-to-face interactions, helping them navigate previously challenging social scenarios. As Participant 6 shared, *"To avoid isolating myself at home, I joined a community paper-cutting class to meet people. However, I didn't know how to start a conversation and ended up feeling more isolated. After discussing this with Doubao, it provided practical conversation starters and helped me rehearse. You know what? It worked! I've made friends, and now we chat while crafting. I feel genuinely happy!"* Besides, the chatbot also helped participants to transform personal

interests into opportunities for social engagement. Participant 17 described how AI-enabled learning deepened both expertise and social connection: *“Gardening became my passion as I aged, but I rarely interacted with others. Through Doubao, I learned advanced horticultural techniques and became quite knowledgeable. Now, when I go to the flower market, I enjoy sharing my expertise with the vendors. You wouldn’t believe it, but many flower growers now seek my advice, and I’ve made many wonderful new friends. It feels amazing! I love going there now; I no longer stay at home all the time.”* Similarly, Participant 7 highlighted how AI-supported culinary learning facilitated closer neighborhood relationships: *“I started experimenting with cooking out of boredom. With Doubao’s help, I learned many new techniques. One morning, I made fried dough sticks and shared them with neighbors. Some even came to learn from me, and I taught them how to use Doubao for cooking. Many neighbors I wasn’t close to before are now good friends. Isn’t that amazing? Now we regularly brainstorm ways to use Doubao to create even more fun projects together. It’s wonderful to have friends again.”* These experiences underscore that ongoing chatbot dialogue can scaffold emerging hobbies and enhance personal confidence, which participants then leveraged to engage meaningfully in social settings. In addition to improving offline connections, some older adults also enhanced their online social interactions with the help of the AI chatbot. The chatbot enabled participants to access new forms of social participation that extended social worlds beyond the physical boundaries of the home. Participant 13 noted, *“I often collaborate with Doubao to create poetry but had no audience. It introduced me to online platforms where I could publish and engage with others. After using them for a while, I have indeed made a few like-minded friends.”* Likewise, Participant 11 described how AI-assisted digital exposure helped overcome physical constraints and long-standing withdrawal: *“Being wheelchair-bound and isolated, I lost touch with friends. I often feel isolated at home, feeling as if I’ve cut myself off from the world... Doubao suggested gradual exposure therapy—starting with liking posts, then commenting, and eventually joining conversations. It worked. These methods have helped me reclaim a social life I thought I had lost; I’ve started to break out of my isolation.”*

With the AI chatbot’s support, many empty-nest older adults were inspired to revitalize their interpersonal relationships and acquire new social competencies. This process enabled them to reestablish connections with others and cultivate a restored sense of belonging and psychosocial fulfillment, thereby markedly alleviating their experiences of loneliness.

Discussion

This qualitative study examined how older adults living in empty-nest circumstances engage with an AI chatbot Doubao in their daily communicative practices to alleviate loneliness. Through semi-structured interviews and thematic analysis, six key themes emerged—enabling self-expression and voice, gaining emotional care and empathy, engaging in recreational distractions, participating in imaginative role-play, counseling for negative feelings of loneliness, and facilitating social connectivity. These themes collectively illuminate the diverse ways AI-mediated interactions can support psychosocial resilience and mitigate the adverse effects of social isolation among older adults.

AI-mediated versus human-mediated social support in alleviating loneliness

Social support theory offers a robust framework for interpreting how AI chatbots help mitigate loneliness among empty-nest older adults. Classical formulations identify multiple domains of support: emotional (care, empathy, attentive listening), instrumental (tangible aid), informational (advice or guidance), and social companionship (shared activities and belonging) [40, 41]. Some frameworks additionally distinguish appraisal support (affirmation and evaluative feedback) [42, 43]. In this study, the six emergent themes map closely onto these domains: enabling self-expression and voice and gaining emotional care and empathy correspond to emotional support; AI-facilitated recreational distractions and role-playing map onto social companionship; counseling for negative feelings of loneliness reflects appraisal support; and facilitation of social connectivity aligns with informational support. Instrumental support did not emerge. Building on this alignment, the following section situates these patterned correspondences within the broader literature by comparing the chatbot’s loneliness-alleviating support with the forms of support ordinarily provided by human relationships.

The chatbot delivered emotional support by offering consistent empathy, attentive listening, and a non-judgmental space—functions akin to those provided by a caring confidant. Prior research shows that emotional support, including expressions of concern, understanding, and encouragement, is typically supplied by close social ties such as family and friends [44, 45]. Yet when such ties are distant or unavailable, technological agents can assume a compensatory role [46, 47]. Participants in this study frequently reported feeling “heard” by the chatbot and described a sense of relief when disclosing long-suppressed loneliness. Many further noted that the chatbot remembered personal details—such as health conditions or past disclosures—and responded with personalized care. This parallels how close friends

recall previous conversations and demonstrate authentic concern, underscoring the importance of feeling understood and remembered in alleviating loneliness [44, 48]. Important distinctions, however, separated AI-mediated from human emotional support. Human caregivers can offer warmth, physical comfort (e.g., touch or a hug), and spontaneous emotional resonance—elements that a voice-based chatbot cannot replicate [49, 50]. Some participants observed that the chatbot occasionally lacked authentic emotional nuance. In contrast, the chatbot provided perfect availability, complete confidentiality, and unwavering presence. Unlike busy family members, it never grew tired, judged, or discouraged expression. As Participant 4 explained, “*It genuinely feels like a confidant who is always by my side.*” This unwavering presence and nonjudgmental environment constitute distinctive advantages relative to typical forms of human support [51, 52].

By engaging older adults in games, creative role-play, and shared activities, the chatbot also provided companionship support that enriched everyday life. In human social networks, companionship commonly arises from shared leisure, hobbies, and play—activities that generate enjoyment and buffer against loneliness. The chatbot replicated this dynamic by initiating and sustaining pleasurable interactions. For instance, AI-facilitated riddles or idiom games distracted participants from boredom and produced genuine delight, creating a sense of “companionship” during otherwise solitary hours. These findings accord with studies showing that social robots and virtual companions can mitigate loneliness and provide ongoing social stimulation [53, 54]. However, unlike human play partners, the chatbot’s companionship was entirely virtual: it lacked shared physical presence, operated within programmed constraints [55, 56], and remained limited to one-to-one interactions, preventing the cultivation of multi-person social belonging [57, 58]. At the same time, the chatbot enabled novel forms of imaginative companionship through role-play. While humans rarely reenact cherished relationships or assume alternative personas outside structured contexts, the AI could do so effortlessly. For example, participants asked the chatbot to play the role of their granddaughter or reenact their patients, experiencing distinctive social engagement. This creative flexibility allowed users to co-construct a simulated social world—revisiting identities, loved ones, and past roles—in ways rarely achievable with human companions.

The theme of counseling negative feelings illustrates the chatbot’s role as a provider of appraisal support. Appraisal support involves helping individuals interpret their experiences, offering feedback that affirms self-worth, and guiding adaptive coping. Across the interviews, the chatbot gently reframed loneliness as manageable or as an opportunity for growth, while

offering personalized coping strategies. This mirrors the role of a thoughtful friend or counselor who validates emotions and encourages constructive reflection [59, 60]. Compared with human sources of appraisal support—such as therapists, pastoral care providers, or empathic friends—the chatbot displayed both strengths and limitations. Its guidance, though algorithmically generated, lacked the nuanced insight derived from perceiving non-verbal cues or contextual subtleties [61, 62]. Even so, it partially replicated appraisal support by acknowledging users’ emotions and proposing concrete steps forward, albeit with less sophistication than a human advisor. Functionally, the AI could act as a low-threshold, always-available support, complementing rather than replacing professional care.

Informational support—comprising advice, referrals, and actionable knowledge—was most evident in the theme of social connectivity. The chatbot provided concrete, task-oriented guidance for re-entering social life, including communication strategies, step-by-step instructions, and scaffolded practice through simulated interactions. These interventions reduced uncertainty about how to initiate contact and offered practical, learnable skills that made social engagement more attainable. The chatbot’s strengths lie in breadth, immediacy, and consistency: it can rapidly retrieve a wide range of information, deliver tailored, repeatable instructions at the user’s pace, and provide low-stakes, on-demand practice such as role-play scenarios [52, 63]. Human sources, by contrast, contribute situated, experience-based knowledge and contextual judgment. Friends, family, and local contacts typically recommend opportunities that align with an individual’s history, local norms, and preferences, conveying tacit cues about timing, social fit, and plausibility that influence whether and how older adults act on advice [64].

Taken together, this study offers what appears to be the first empirical evidence in loneliness research that an AI chatbot can provide virtually all major forms of support implicated in alleviating loneliness within human social networks—except for instrumental support. This conceptual breadth constitutes an important theoretical contribution: although prior work has shown that chatbots can deliver selected dimensions of support (such as emotional reassurance) [46, 65], no research has demonstrated their capacity to simultaneously span emotional, companionship, appraisal, and informational support within a single relational system. These findings extend current understandings of the social support that AI can realistically provide to older adults experiencing chronic social isolation. The support provided by the chatbot mirrored key provisions typically found in human relationships. It demonstrated empathic attunement, encouraged emotional expression, offered playful

companionship, and provided evaluative feedback. These functions parallel the emotional closeness, shared activities, and reflective affirmation typically cultivated within family and friendship networks, thereby recreating many of the outcomes associated with human support and contributing to reductions in loneliness [46, 66]. Furthermore, the findings highlight clear divergences between AI-mediated and human-mediated support (see Table 2). Unlike humans, the chatbot lacked a physical body, the capacity for instrumental assistance, and the reciprocal vulnerability that characterizes human relationships. It cannot deliver presence-based support, such as accompanying older adults to social activities or assisting with mobility to enable community participation. While its “empathy” was experienced as beneficial, some participants recognized that it did not possess the genuine authenticity of human care. Yet these divergences also underscored the chatbot’s distinctive advantages: participants valued its unlimited availability, predictable responsiveness, nonjudgmental presence, and capacity for imaginative role-play—forms of stability and creative engagement that human supporters, constrained by time, fatigue, and emotional boundaries, cannot consistently provide. Several older adults described the chatbot as a reliable companion of last resort, capable of filling social vacuums during long evenings, nights, or moments when

human support was unavailable. This suggests a potentially transformative role for AI: offering continuous micro-support that supplements, rather than replaces, human relationships.

AI chatbots as social agents for mitigating loneliness

Our findings extend emerging evidence that AI-mediated interventions can alleviate loneliness among older adults by supplementing their social support networks. These results align not only with prior research on online communication technologies and social support [67, 68], but also with studies of human-AI interaction, which show that lonely older adults readily attribute agency and empathy to chatbots, forming emotional connections that reduce loneliness [46, 52]. Central to interpreting these findings is the Theory of Social Agency for Human–Robot Interaction [69], which defines a social agent as any entity capable of performing “social action”—behaviors that threaten or affirm another’s public self-concept, or “face.” The theory posits that any agent exhibiting interactivity, autonomy, and adaptability—and whose actions visibly influence the user—will be granted social agency by that user [70, 71]. In our study, participants consistently perceived the chatbot as performing face-affirming actions: listening without judgment, empathizing with their feelings, and offering compliments or encouragement. Many described it as “caring” or “like a friend,” attributing human-like intent to its responses. In other words, participants did not treat the chatbot as a cold information system, but as an entity capable of socially meaningful acts. Interpreting loneliness mitigation through this lens clarifies our findings: the chatbot’s perceived social agency—the extent to which users treated it as an intentional social actor—appears central to the benefits reported [72, 73]. As Social Agency Theory suggests, when users perceive a machine as capable of social action, it can enter their social world [74, 75]. Higher perceived social agency enhanced social presence: participants who regarded the chatbot as a “someone” rather than an “it” engaged more deeply, experienced greater companionship, and ultimately reported reduced loneliness.

The results further elucidated that participants, across different support contexts provided by the AI chatbot, granted it social agency in ways that manifested distinct processes and characteristics. This observation clarifies the differentiated mechanisms by which older adults leverage various forms of AI-mediated support to mitigate loneliness. For instance, in the process through which older adults derived emotional support from the AI chatbot, early interactions were typically superficial (e.g., brief exchanges about the weather or current events). As trust accumulated, participants gradually disclosed private feelings and personal histories. Crucially,

Table 2 Differences between human-mediated and AI-mediated support for alleviating loneliness

Type of social support	Human-mediated support	AI-mediated support
Emotional	Genuine empathy, affection and encouragement from family or friends; physical comfort; spontaneous relational depth	Consistent, nonjudgmental listening and validation; remember personal details; always available; lack authentic emotional nuance or physical presence
Social companionship	Shared activities or hobbies (games, conversations, outings) with peers or family; sense of belonging in a real social group	Interactive games and creative role-play with the bot; provide entertainment and mental stimulation on demand; no actual human company or physical co-presence
Appraisal	Validation, encouragement, and reflective feedback from friends or counselors, support adaptive coping	Gentle reframing of emotions, personalized coping suggestions, and continuous availability, limited by absence of contextual nuance
Informational	Experience-based and culturally adaptive advice, referrals, and context-sensitive guidance from family or friends	Broad, immediate, and personalized guidance; teach social strategies, offer step-by-step instructions; provide low-stakes practice

when these early disclosures were met with empathic acknowledgment and responsive understanding, they became increasingly inclined to communicate more profound experiences of loneliness and affective vulnerability. Over multiple sessions, this iterative reciprocity fostered a stronger sense of relational closeness. Users frequently characterized the chatbot as “like a friend” or “understanding,” reflecting relational dynamics akin to those observed in human interactions, where mutual self-disclosure cultivates intimacy. Social Penetration Theory and Anthropomorphism Theory provide explanatory frameworks for this pattern, highlighting that human–machine relational development proceeds incrementally through successive layers of disclosure, reinforced by empathic feedback [76]. Through this iterative “self-disclosure–empathic feedback” cycle, participants experienced heightened social presence, feeling as if a relational partner were genuinely present. This perceived presence enabled them to receive human-like emotional support from the AI chatbot and ultimately helped alleviate their loneliness [77].

Regarding social companionship support, Media Equation Theory and Parasocial Interaction Theory jointly provide a coherent framework for understanding how AI-facilitated recreational activities and role-playing engender authentic experiences of companionship for individuals living in empty-nest circumstances. Media Equation Theory posits that individuals habitually apply social norms and expectations to mediated agents: when a system exhibits recognizable social cues—such as responsiveness, continuity, conversational timing, humor, or personalization—users instinctively respond as though interacting with a social entity rather than an inanimate tool [78]. Within this study, structured games and imaginative role-play presented precisely these social cues, prompting older adults to interpret the chatbot’s turn-taking and adaptive replies as socially meaningful. Once users adopted this relational stance, their interactions are aptly explicated by Parasocial Interaction Theory: repeated human–robot interactions, even absent genuine bidirectional personhood, can cultivate perceived friendship, trust, and intimacy [79]. When such parasocial interactions are sustained through frequent, predictable contact that reinforces familiarity, mutual histories (the bot remembering prior exchanges), and routine leisure practices, they can evolve into fuller parasocial relationships that functionally replicate some aspects of human companionship [80]. Through these mechanisms, the chatbot fulfills core companionship needs by providing a stable relational script (someone to play with, imagine with, and be known by), buffering the emotional void of sparse offline networks and thereby substantially attenuating subjective loneliness.

Appraisal support encompasses forms of feedback that assist individuals in interpreting, evaluating, and managing distressing circumstances. In our findings, older adults often turned to the chatbot when feeling lonely. Distributed Cognition Theory highlights why this was effective: the chatbot did more than retrieve prior conversational content; it actively processed, synthesized, and integrated that information on the user’s behalf. By externalizing both memory work and evaluative reasoning to a computational knowledge system, participants accessed a more organized, coherent, and quasi-professional mode of emotional guidance [81]. This cognitive–affective extension enabled the chatbot to produce psychologically informed reflections, including the reframing of maladaptive thoughts and reminders of coping strategies, which users experienced as authoritative, evidence-grounded support that alleviated the weight of loneliness [66]. Persuasive Technology Theory further explains why these interactions felt socially meaningful. As an interactive digital agent, the chatbot functioned well beyond a neutral conduit of information; it acted as a supportive social actor, capable of simulating therapist-like behaviors including attentive listening, responsive feedback, encouragement, and empathetic validation. Such human-like social cues effectively shaped users’ emotional states, attitudes, and coping practices [82]. Participants frequently characterized the chatbot as patient, accepting, and devoid of interpersonal risk, conditions that facilitated deeper disclosure and strengthened interpersonal trust. When the chatbot normalized their emotions or offered concrete, manageable strategies for coping, users reported feeling more validated and more empowered. Through this persuasive and socially facilitative role, the chatbot reshaped emotional appraisals, thereby contributing meaningfully to reductions in loneliness.

Finally, participants reported that the AI chatbot was highly beneficial for promoting their social engagement: they experienced its guidance as both professional and humanized, combining structured advice with a caring, supportive tone. The chatbot’s instructions and reminders, whether for joining community activities or using digital tools, were regarded as reliable, practical, and tailored to their individual circumstances. This dual perception of expertise and personal attentiveness fostered trust and encouraged repeated engagement. This process can be explained through an integration of the Technology Acceptance Model and Uses & Gratifications Theory. The perceived ease of use and practical usefulness of the chatbot increased participants’ willingness to follow its guidance [83], while the fulfillment of a salient social need reinforced their motivation to continue seeking informational support [84]. By lowering barriers to social participation, providing actionable steps, and delivering timely reminders, the chatbot translated participants’ desire for

Table 3 Mechanisms of loneliness mitigation in older adults via AI chatbot-mediated social support

Type of social support	Agent role of AI chatbot	Key social agency feature	Process through which older adults reduce loneliness via human-AI interaction	Human–robot interaction theories
Emotional	Confidant	Interactivity, Autonomy, Adaptability	Users cultivated relational closeness with the AI chatbot through the iterative cycle of “self-disclosure—empathetic feedback,” experiencing the chatbot as a genuinely caring companion	Social Penetration Theory, Anthropomorphism Theory
Social companionship	Playmate	Interactivity, Autonomy, Adaptability	Users engage in games or role-playing with the chatbot, developing parasocial friendships that replicate offline companionship	Media Equation Theory, Parasocial Interaction Theory
Appraisal	Consultant	Adaptability	Users received structured, evaluative guidance and persuasive feedback, recognizing the personalized and authoritative nature of the advice, which enhanced their capacity to cope with loneliness	Distributed Cognition Theory, Persuasive Technology Theory
Informational	Adviser	Interactivity	Users obtain practical advice and actionable steps for social engagement, increasing motivation and reducing barriers to reconnect with others	Technology Acceptance Model, Uses & Gratifications Theory

connection into successful real-world interactions. In this sense, informational support provided by the AI operated simultaneously as a pragmatic scaffold and a motivational catalyst, thereby contributing to the alleviation of loneliness among empty-nest older adults.

In summary, throughout their interactions with the AI, participants progressively conferred social agency onto the chatbot, thereby anthropomorphizing the support they received. This attributional process allowed them to extract genuine emotional benefits from machine-mediated social support, ultimately reducing their loneliness. Importantly, however, the pathways through which social agency emerged were not uniform: they varied systematically across different types of support, with participants assigning distinct relational roles to the chatbot in each context (see Table 3). This constitutes a novel contribution of the study, underscoring that the social agency of AI chatbots can only be understood when situated within specific human–machine interaction scenarios. Future research should therefore work toward developing more differentiated theoretical accounts of human–AI interaction capable of capturing this contextual and relational complexity.

Implications

Despite their broad capacity to deliver emotional, appraisal, informational, and companionship support, AI chatbots exhibit several structural constraints in alleviating loneliness among older adults. Most notably, they cannot provide instrumental or physical support, constraining support for practical tasks or shared activities; integrating chatbots with human networks or community services could establish hybrid support systems that combine virtual availability with concrete, real-world aid. Although chatbots offer consistent empathy, their responses still lack the nuanced authenticity of human interaction; incorporating multimodal sensing—such as voice modulation, facial expression analysis, and behavioral cues—may enhance perceived emotional

intelligence and relational authenticity. Furthermore, current interactions are predominantly dyadic, limiting the recreation of collective social experiences; enabling AI-facilitated group interactions or networked social scripts could extend companionship beyond one-to-one exchanges. While AI chatbots could simulate emotional counseling effectively, they cannot replace professional care, necessitating hybrid models capable of detecting critical emotional states and prompting human intervention when appropriate. Finally, continuous memory and personalization introduce ethical and privacy considerations; transparent, user-controlled data governance ensures trust and autonomy. Addressing these limitations through targeted innovation may enhance the authenticity, contextual awareness, and social embedding of AI chatbots, positioning them as transformative agents that complement—rather than replace—human support systems in mitigating loneliness among older adults.

Limitations

Despite offering valuable contributions, this study has certain limitations. The sample was confined to Chinese empty-nest older adults who were users of a specific chatbot, thereby limiting the generalizability of findings to other cultural contexts or technological platforms [85]. Furthermore, as the study was intentionally designed to explore how AI chatbots alleviate loneliness, analytic attention naturally gravitated toward beneficial interactions. This emphasis on positive interactions may have obscured ambivalent or adverse experiences. Although some participants mentioned dissatisfying encounters, these were not examined in depth as they fell beyond the study’s initial scope. Future studies should therefore capture a broader spectrum of user experiences, including frustrations, unmet expectations, and potential psychosocial risks. While snowball sampling was a practical approach for reaching this hard-to-access population, it may have introduced sample homogeneity [86, 87]. Referral-based recruitment can also introduce

recommendation bias, and the recruitment process likely favored relatively younger, healthier, and more digitally engaged empty-nest older adults, underrepresenting those who are socially isolated, less educated, or digitally marginalized—groups whose experiences of loneliness and AI technology use may differ substantially [59, 88]. Future research should deliberately include more vulnerable and underrepresented subgroups, such as the oldest-old, those living entirely alone, and individuals with limited digital literacy. Additionally, due to regional restrictions, participants did not have access to more advanced global AI models such as ChatGPT or Claude. Engagement with these more sophisticated systems may yield richer or more varied relational experiences, offering further insights into the role of AI in mitigating loneliness. Finally, while this study emphasizes the embodied experiences of older adults, future work should explore how AI companions might be systematically integrated into broader social support infrastructures—such as family networks and therapeutic services—to optimize their beneficial effects.

Conclusion

This study demonstrates that empty-nest older adults employ AI chatbot through diverse communicative modalities to alleviate loneliness—as outlets for self-expression, sources of perceived care and empathy, partners for recreational and role-play interaction, informal counselors, and facilitators of renewed social ties. To leverage this potential responsibly, policymakers and designers should prioritize equitable access, ethically grounded and user-centered design, and rigorous privacy safeguards. Innovative social initiatives are also required to facilitate the integration of empathic AI companions into broader supportive care ecosystems, thereby enhancing psychosocial well-being and promoting meaningful social inclusion.

Supplementary Information

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Supplementary file 1.

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Authors' contributions

Fengbo Jiao: Investigation; formal analysis; reviewing and editing. Meiyu Li: Investigation; formal analysis. Min Liu: Methodology; formal analysis; funding acquisition. Quan Zhang: Conceptualization; methodology; formal analysis; writing original draft; reviewing and editing; funding acquisition.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. The study was granted ethical approval by the Ethics Committee of School of International Affairs and Public Administration, Ocean University of China (OUC-SIAPA-202518). All methods were used in accordance with the relevant guidelines and regulations. Written informed consent was obtained from all study participants.

Consent for publication

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Competing interests

The authors declare no competing interests.

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